PORTFOLIO CONSTRUCTION

Inverse-Volatility Weighting with a Speculative Micro-Cap Sleeve

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Research period: 2 February 2023 to 6 November 2026

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Headline result

The portfolio produced a full-period Sharpe ratio of 1.224 versus 0.920 for VOO and a positive HAC-adjusted alpha. However, performance is positively skewed and materially influenced by the speculative TIPS equity position.

Goal of the portfolio:

The portfolio construction study adopts inverse-volatility weighting to control risk. If an asset has higher volatility, a lower weight will be assigned to it. A sudden price jump may be followed by:

- profit-taking;
- a price reversal;
- wider bid-ask spreads;
- unstable subsequent returns.

Reducing the weight can therefore limit losses if the price increase proves temporary.

Main weakness

If an asset rises strongly and continues to appreciate, inverse-volatility weighting may reduce the position too early and sacrifice further upside.

Executive Summary

Metric	Joe's Portfolio	VOO	Interpretation
Annualized excess return	20.15%	13.47%	Portfolio earned 6.68 percentage points more per year
Sharpe ratio	1.224	0.920	Higher historical excess return per unit of volatility
VOO beta	1.007	1.000	Market sensitivity was approximately equal to VOO
Daily alpha	≈0.03%	—	Positive estimate; exact annualization requires the unrounded coefficient
Alpha p-value	0.031	—	Statistically significant at 5%, but economically fragile

The strategy uses trailing 20-day inverse-volatility weights and applies a one-day signal lag. It therefore reduces exposure to recently volatile assets and increases exposure to recently stable assets. The portfolio is equity-heavy rather than a conventional multi-asset portfolio.

- 2023: the portfolio underperformed VOO because concentrated U.S. large-cap leadership was stronger than the diversified allocation.
- 2024: the portfolio approximately matched VOO on a risk-adjusted basis.
- 2025 and 2026: the portfolio materially outperformed VOO, with TIPS contributing the most annualized return in both periods.
- Key caution: TIPS is Tianrong Internet Products and Services, Inc., an OTC speculative equity. It is not Treasury Inflation-Protected Securities.

1. Investment Objective and Design

The objective is to evaluate whether a dynamically weighted equity portfolio can generate higher risk-adjusted returns than a passive investment in VOO. The portfolio combines broad U.S. exposure, growth exposure, small-cap exposure, developed international exposure, and a speculative micro-cap sleeve.

Ticker	Exposure	Role	Primary risk
QQQ	Nasdaq-100 growth equities	Growth and technology tilt	Concentration and valuation risk
VOO	S&P 500 large-cap equities	Core U.S. market exposure	Broad U.S. equity drawdowns
IWM	Russell 2000 small caps	Size diversification	Credit, funding and economic sensitivity
EFA	Developed markets ex U.S./Canada	Geographic diversification	Currency and regional-market risk
TIPS	Tianrong Internet Products and Services, Inc.	Speculative tail-return sleeve	OTC liquidity, extreme volatility and execution risk

The portfolio should therefore be interpreted as a growth-oriented global equity strategy with a speculative micro-cap component, not as a balanced stock-bond portfolio.

2. Methodology

2.1 Inverse-volatility weighting

$$w_{i,t} = \frac{1/\sigma_{i,t}}{\sum_{j=1}^{N} 1/\sigma_{j,t}}$$

where $\sigma_{i,t}$ is the annualized standard deviation of the asset's previous 20 daily returns. Lower-volatility assets receive larger weights.

Portfolio return uses lagged weights:

$$R_{p,t} = \sum_{i=1}^{N} w_{i,t-1} R_{i,t}$$

- The one-day lag avoids look-ahead bias.
- Weights are normalized to 100% each day.
- The method uses individual volatilities but not the covariance matrix, so it is not full equal-risk-contribution risk parity.
- Daily rebalancing is assumed. Transaction costs, bid-ask spreads, market impact and partial fills are not deducted.

Illustrative portfolio weights

The following five trading days provide a worked example of the normalized inverse-volatility weights produced by the model. Each row sums to approximately 100%, subject to rounding.

Date	QQQ	VOO	IWM	EFA	TIPS
2026-11-02	15.01%	29.61%	20.89%	33.61%	0.88%

2026-11-03	14.95%	30.04%	21.10%	33.03%	0.88%
2026-11-04	16.51%	29.68%	23.70%	29.07%	1.04%
2026-11-05	16.98%	26.74%	23.02%	31.79%	1.46%
2026-11-06	16.73%	27.05%	22.88%	31.87%	1.47%

Table 2. Example inverse-volatility portfolio weights, 2-6 November 2026.

For example, on 6 November 2026 the model allocated 31.87% to EFA, 27.05% to VOO, 22.88% to IWM, 16.73% to QQQ and 1.47% to TIPS. The small TIPS allocation reflects its substantially higher trailing volatility.

3. Diversification and Correlation

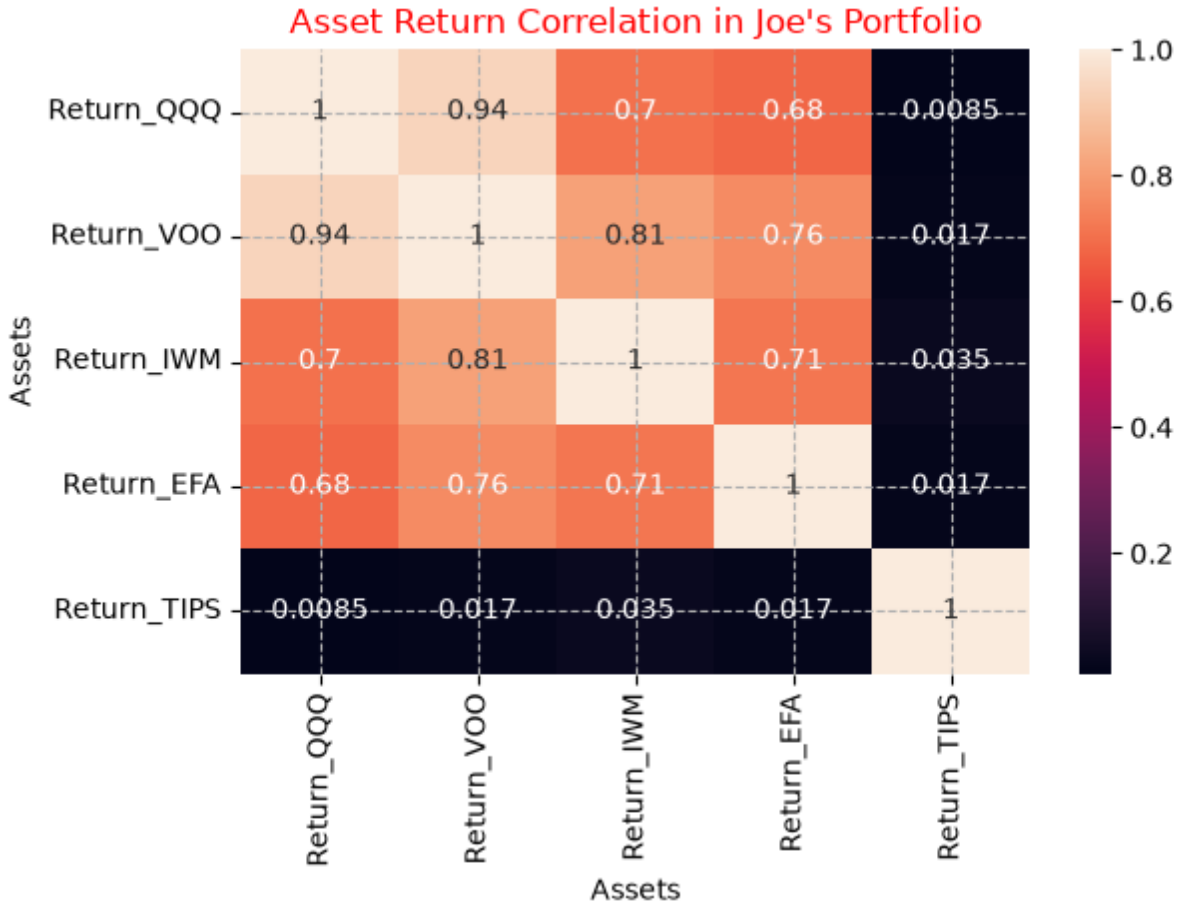
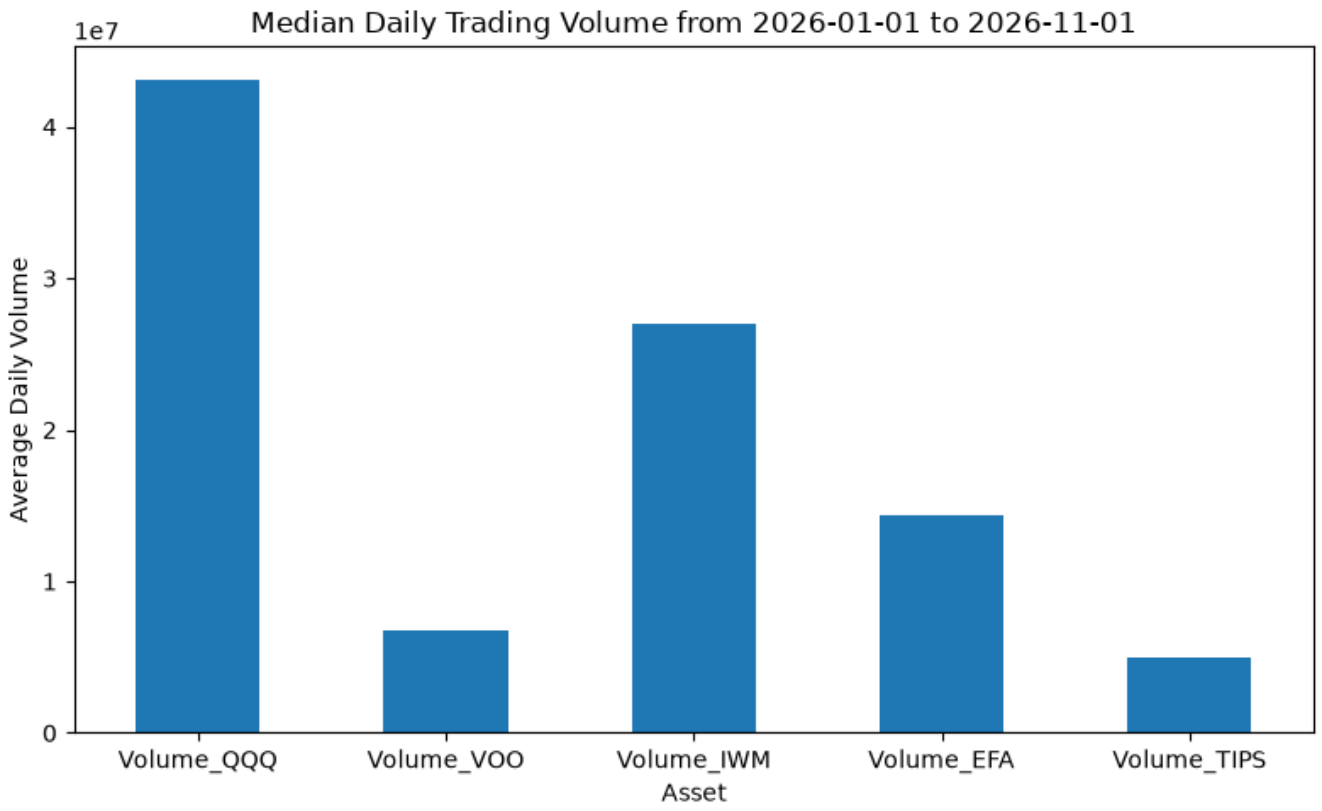


Figure 1. Daily return correlations, 2 February 2023 to 6 November 2026.

The four conventional equity exposures are highly correlated, with correlations ranging from approximately 0.68 to 0.94. QQQ and VOO are especially similar, at 0.94. TIPS has near-zero measured correlation with the conventional equity exposures, but this apparent diversification benefit must be interpreted cautiously because the potential illiquidity of TIPS may offset the benefit of the diversification if it is not tradable. The following liquidity analysis therefore examines trading activity separately.



The chart compares median daily share volume from 1 January to 1 November 2026. TIPS recorded regular trading activity, with median share volume broadly comparable to VOO. Accordingly, the chart supports good TIPS liquidity.

4. Annual Return Contribution

Daily asset contribution is calculated as the prior-day weight multiplied by the current-day asset return. The charts below annualize the arithmetic mean of daily contributions. These contributions explain average portfolio return, but they are not an exact compounded wealth attribution.

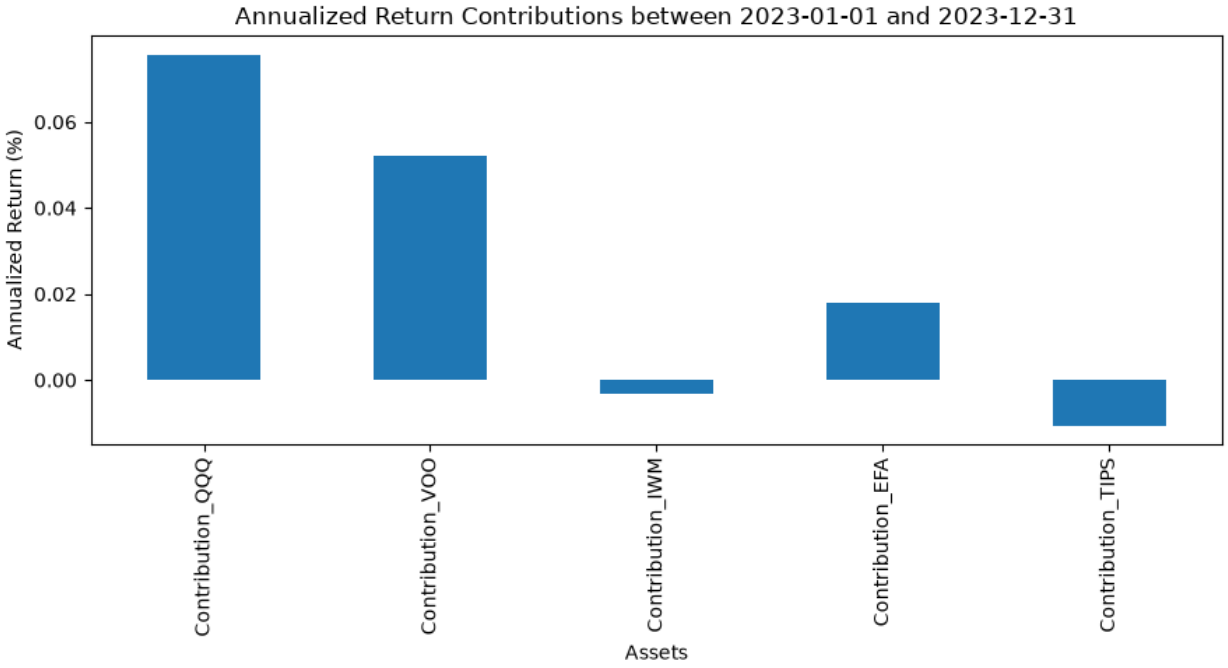


Figure 2. Annualized arithmetic return contribution, 2023.

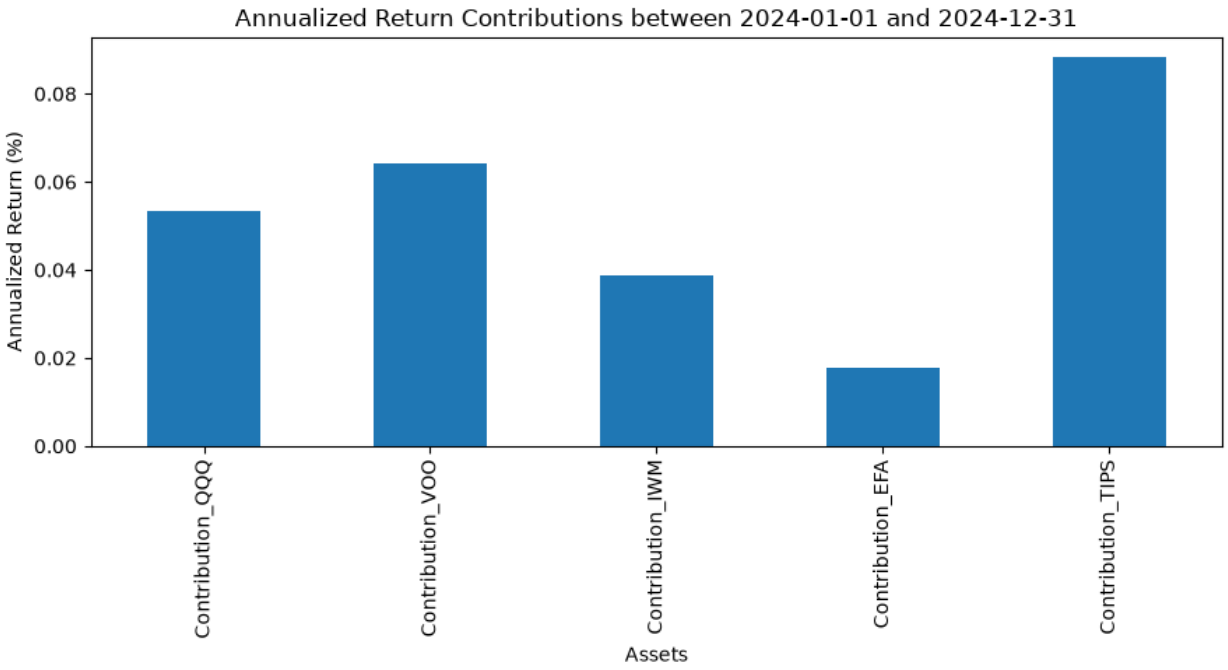


Figure 3. Annualized arithmetic return contribution, 2024.

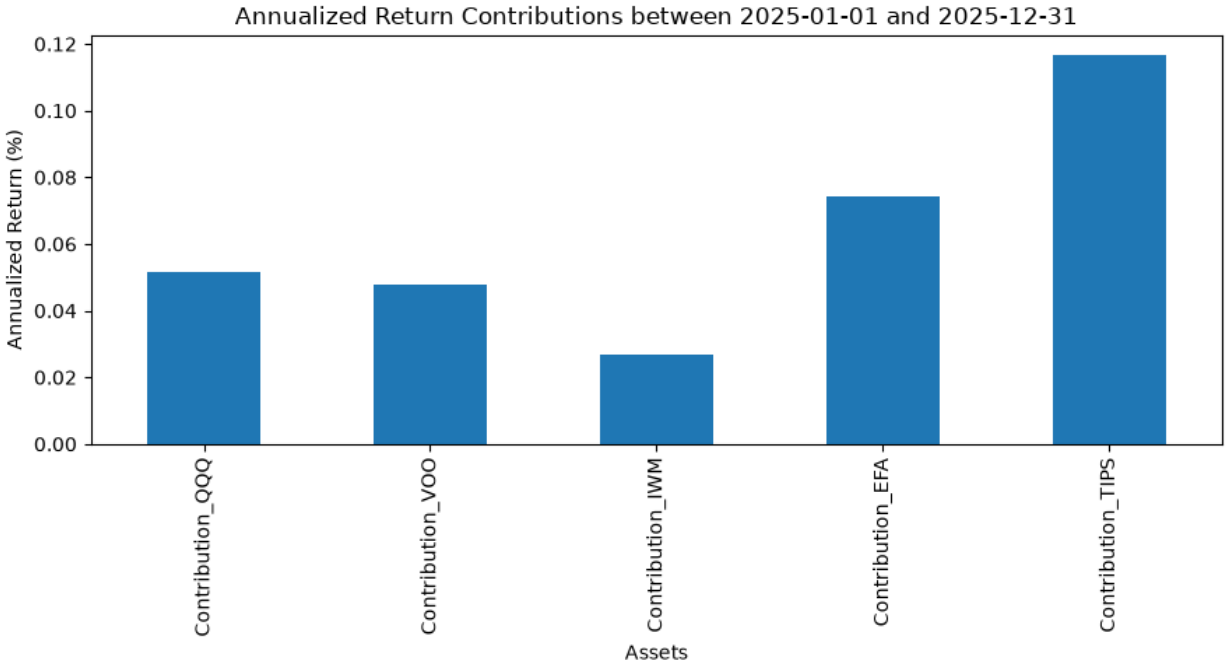


Figure 4. Annualized arithmetic return contribution, 2025.

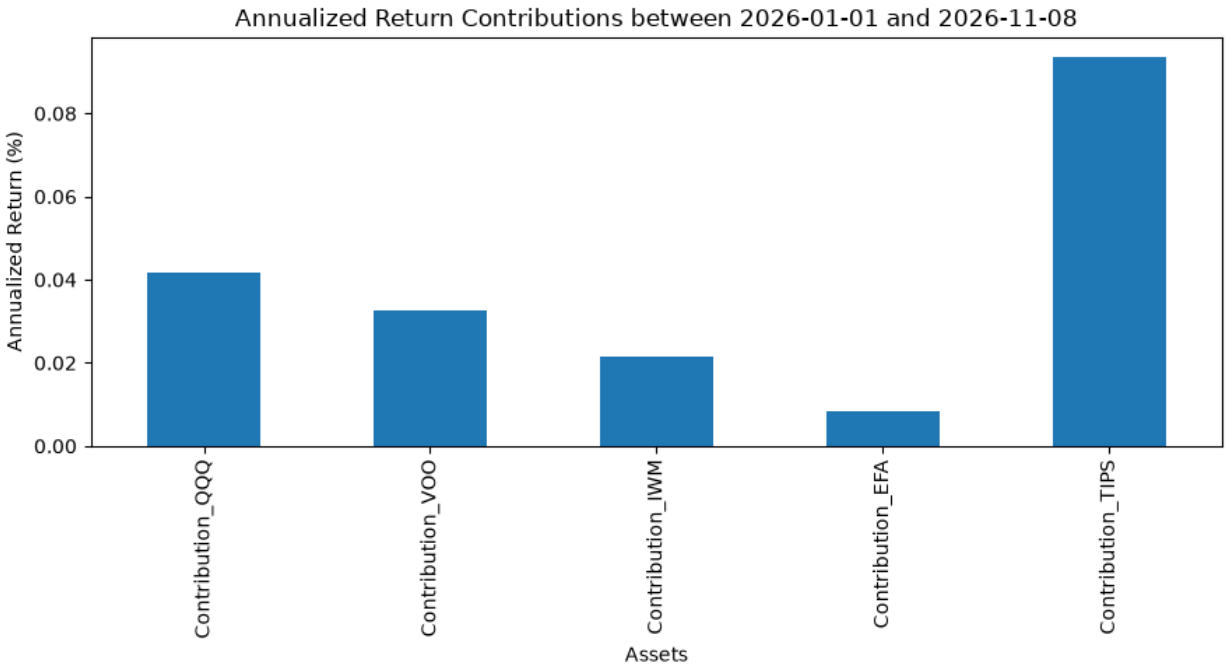


Figure 5. Annualized arithmetic return contribution, 2026 to 6 November.

TIPS was the largest contributor in 2024, 2025 and 2026 despite usually receiving the smallest weight. This occurs because percentage changes can be very large when an OTC penny stock moves from one very small quoted price to another. A small position can therefore have a material portfolio effect. The portfolio outperformed VOO largely because of the contribution from TIPS.

5. Performance by Test Period

Period	Portfolio Sharpe	VOO Sharpe	Portfolio excess	VOO excess	Assessment

			return	return	
2023-02-02 to 2023-12-29	0.638	1.066	9.54%	13.51%	Underperformed
2024-02-01 to 2024-12-30	1.495	1.491	22.44%	19.01%	Approximately matched
2025-02-04 to 2025-12-30	1.459	0.763	28.67%	14.46%	Strong outperformance
2026-02-03 to 2026-11-06	0.886	0.397	15.26%	5.61%	Outperformed
Full period	1.224	0.920	20.15%	13.47%	Outperformed overall

Portfolio Growth versus VOO

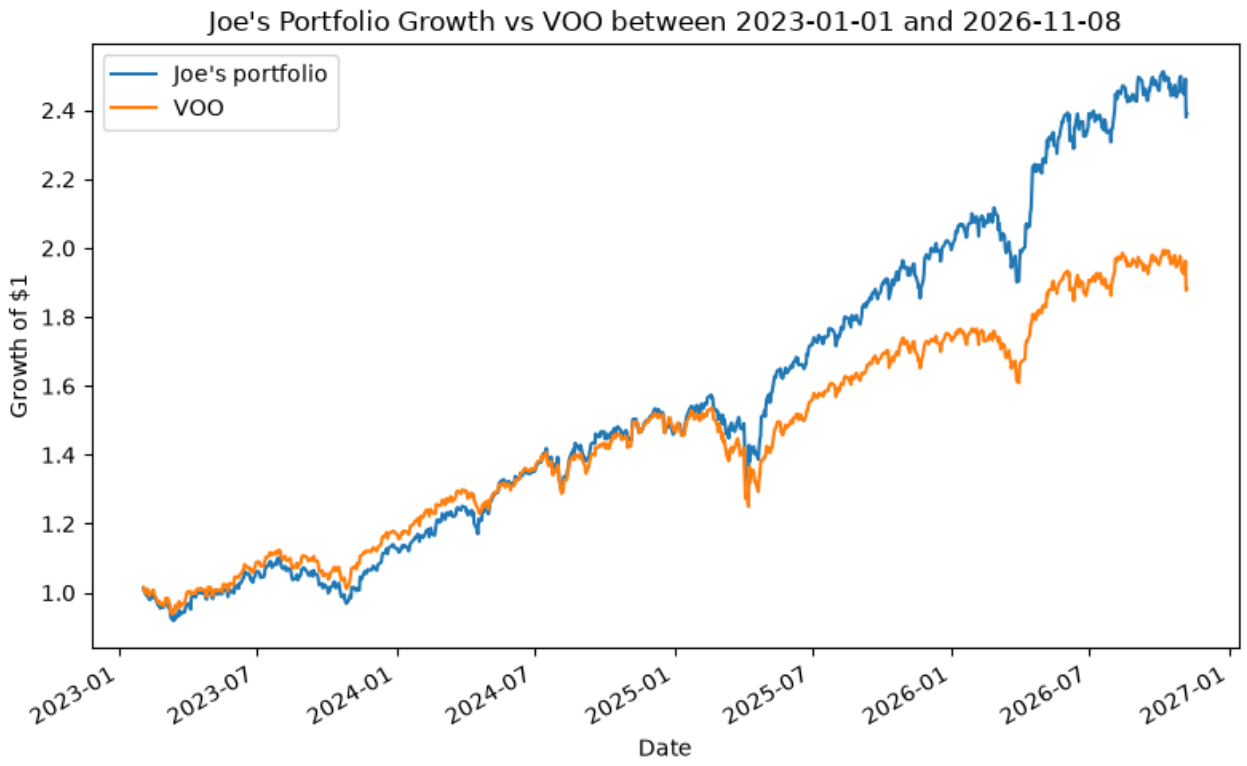


Figure 6. Growth of US$1 invested in Joe's portfolio and VOO, 2 February 2023 to 6 November 2026.

The portfolio and VOO followed broadly similar market cycles through early 2025, after which the portfolio's cumulative growth separated materially from the benchmark. The wider gap reflects stronger portfolio returns in 2025 and 2026, but should be interpreted together with the contribution analysis because the speculative TIPS sleeve materially influenced later performance.

Economic interpretation

In 2023, VOO benefited from concentrated U.S. mega-cap leadership. The inverse-volatility rule allocated part of the portfolio to small-cap, international and speculative exposures, producing lower return and higher volatility than VOO. In 2024, higher portfolio return was largely offset by higher risk, leaving almost no Sharpe advantage. In 2025 and 2026,

non-VOO holdings generated enough additional return to compensate for higher volatility, while TIPS provided the largest annualized contribution.

6. Benchmark Regression and Alpha

The HAC-robust benchmark regression is:

$$R_{p,t} - R_{f,t} = 0.0003 + 1.0071\left(R_{\text{VOO},t} - R_{f,t}\right) + \varepsilon_t$$

Statistic	Estimate	Interpretation
Alpha (daily)	≈0.0003	Positive abnormal return estimate
Alpha z-statistic	2.163	2.163 robust standard errors above zero
Alpha p-value	0.031	Reject α = 0 at the 5% level
VOO beta	1.0071	Market sensitivity approximately equal to VOO
R-squared	0.802	VOO explains 80.2% of daily portfolio variation
Observations	944	Approximately 3.75 trading years

The estimated alpha is statistically significant at the 5% level. However, statistical significance does not establish economic repeatability. Residuals are strongly non-normal, with skewness of 1.903 and kurtosis of 18.075, consistent with extreme positive tail events. The alpha should therefore be reported alongside robustness tests that exclude TIPS, cap its weight, incorporate realistic transaction costs, and isolate TIPS-driven return days.

7. Suitable Investor Profile

This portfolio is most suitable for **experienced, growth-oriented investors** with a high tolerance for volatility, a long-term investment horizon of at least five years, and no immediate liquidity needs. Investors must be comfortable with substantial equity-market exposure, dynamic portfolio weights, and a small speculative OTC equity position. It is unsuitable for conservative, income-dependent, or capital-preservation investors.

8. Overall Assessment

The portfolio achieved stronger full-period historical performance than VOO, with a Sharpe ratio of 1.224 versus 0.920 and a positive HAC-adjusted alpha. The strategy also demonstrated that lagged inverse-volatility weights can adapt to changing measured risk without using contemporaneous returns.

The evidence is nevertheless mixed across periods. The portfolio underperformed VOO in 2023, approximately matched it in 2024, and outperformed materially in 2025 and 2026. The later advantage coincided with substantial TIPS contributions. Consequently, the strategy combines a conventional equity core with a speculative, positively skewed sleeve rather than delivering universally superior risk control.